IE 400 – Term Project Report

Spring 2026

Question 1: Museum Escape

Problem Definition

Lara starts at cell (1, 1) and must reach the exit at cell (8, 8) on an 8×8 grid in the minimum number of time steps. At each time step she may move one cell up, down, left, or right, or stay in place. She cannot enter obstacle cells. Three security cameras patrol the museum; if Lara occupies any camera-visible cell at the same time step, she is caught.

Obstacle cells: (1, 3), (2, 2), (2, 5), (3, 7), (4, 2), (4, 3), (5, 5), (6, 4), (7, 1), (7, 4), (7, 5)

Camera Movement Tables

Camera 1:

t	Position	Direction	Visible Cells
0	(3, 3)	Right	(3, 3), (3, 4), (3, 5)
1	(3, 4)	Right	(3, 4), (3, 5), (3, 6)
2	(3, 5)	Left	(3, 5), (3, 4), (3, 3)
3	(3, 4)	Left	(3, 4), (3, 3), (3, 2)
4+	repeats from $t = 0$		

Camera 2:

t	Position	Direction	Visible Cells
0	(4, 6)	Down	(4, 6), (5, 6)
1	(5, 6)	Down	(5, 6), (6, 6)
2	(6, 6)	Right	(6, 6), (6, 7)
3	(6, 7)	Left	(6, 7), (6, 6)
4	(6, 6)	Up	(6, 6), (5, 6)
5	(5, 6)	Up	(5, 6), (4, 6)
6+	repeats from $t = 0$		

Camera 3:

t	Position	Direction	Visible Cells
0	(7, 7)	NE	(7, 7), (6, 8)
1	(6, 8)	SW	(6, 8), (7, 7), (8, 6)
2	(7, 7)	SW	(7, 7), (8, 6)
3	(8, 6)	SW	(8, 6), (7, 7), (6, 8)
4+	repeats from $t = 0$		

The combined camera-visible set at time t is

$$F(t) = C_1[t \bmod 4] \cup C_2[t \bmod 6] \cup C_3[t \bmod 4].$$

Task 1: Integer Programming Formulation

Sets and Parameters

- V: set of all non-obstacle cells (r, c), where $r, c \in \{1, \ldots, 8\}$
- $N(r, c)$: set of cells reachable in one step from (r, c), including (r, c) itself and its orthogonal neighbours in V
- $F(t)$: set of cells visible by any camera at time t
- T: planning horizon (total number of steps); to be minimised

Decision Variables

$$x_{t,r,c} \in \{0, 1\} \qquad t = 0, 1, \ldots, T, \quad (r, c) \in V$$

$x_{t,r,c} = 1$ if Lara occupies cell (r, c) at time step t, and 0 otherwise.

Objective Function

$$\min\ T$$

Constraints

(C1) Initial position:

$$x_{0,1,1} = 1, \qquad x_{0,r,c} = 0 \quad \forall\, (r, c) \in V,\ (r, c) \neq (1, 1)$$

(C2) One cell per time step:

$$\sum_{(r,c)\in V} x_{t,r,c} = 1 \qquad \forall\, t = 0, 1, \ldots, T$$

(C3) Valid movement:

$$x_{t+1,r,c} \leq \sum_{(r',c')\in N(r,c)} x_{t,r',c'} \qquad \forall\, t = 0, \ldots, T-1, \quad (r, c) \in V$$

(C4) Camera avoidance:

$$x_{t,r,c} = 0 \qquad \forall\, t, \quad \forall\, (r, c) \in F(t), \quad (r, c) \neq (8, 8)$$

The exit cell $(8, 8)$ is exempt: Lara may step there even if a camera sees it.

(C5) Exit reached at horizon T:

$$x_{T,8,8} = 1$$

(C6) Binary domain: $x_{t,r,c} \in \{0, 1\}$

Task 2: Base Case

BFS over the time-expanded state space finds $T^* = 14$. The IP is solved at $T = 14$ with Gurobi and all constraints (C1)–(C6) are verified to hold.

Result: $T^* = 14$ steps.

t	Cell	t	Cell	t	Cell
0	(1, 1)	5	(5, 2)	10	(8, 4)
1	(2, 1)	6	(6, 2)	11	(8, 5)
2	(3, 1)	7	(7, 2)	12	(8, 6)
3	(4, 1)	8	(8, 2)	13	(8, 7)
4	(5, 1)	9	(8, 3)	14	(8, 8)

Lara goes straight down column 1 to row 5, cuts across to column 2, descends to row 8, then moves right to the exit. No camera is active along this path at the relevant time steps.

Task 3: Booby Trap

Additional rule: if Lara visits cell (5, 1), she may never visit cell (8, 3) afterwards.

We introduce a cumulative binary variable $y_t \in \{0, 1\}$, where $y_t = 1$ means Lara has visited (5, 1) at or before time t. The linearisation is:

$$
\begin{aligned}
y_t &\geq x_{t,5,1} && \forall\, t && \text{(trap triggered)} \\
y_t &\geq y_{t-1} && \forall\, t \geq 1 && \text{(stays triggered)} \\
x_{t,8,3} &\leq 1 - y_{t-1} && \forall\, t \geq 1 && \text{(block (8, 3) after trap)}
\end{aligned}
$$

Result: $T^* = 16$ steps. The base-case path visits (5, 1) at $t = 4$, which would then forbid (8, 3). The solver finds an alternative route avoiding (8, 3) at the cost of 2 extra steps.

t	Cell	t	Cell	t	Cell
0	(1, 1)	6	(5, 3)	12	(5, 7)
1	(2, 1)	7	(5, 4)	13	(5, 8)
2	(3, 1)	8	(4, 4)	14	(6, 8)
3	(4, 1)	9	(4, 5)	15	(7, 8)
4	(5, 1) *[trap]*	10	(4, 6)	16	(8, 8)
5	(5, 2)	11	(4, 7)		

Task 4: Exit Lock ($t < 18$)

Additional rule: the cells LOCKED = {(6, 7), (7, 6), (7, 7), (8, 6), (6, 8)} are forbidden before $t = 18$:

$$x_{t,r,c} = 0 \qquad \forall\, t < 18, \quad (r, c) \in \text{LOCKED}.$$

Result: $T^* = 20$ steps. Lara advances towards the center immediately and spends extra time moving back-and-forth or waiting at safe cells like (4, 4) and (4, 7) to ensure she arrives at (6, 8) exactly at t=18.

t	Cell	t	Cell	t	Cell
0	(1, 1)	8	(5, 3)	16	(5, 7)
1	(2, 1)	9	(5, 4)	17	(5, 8)
2	(3, 1)	10	(4, 4)	18	(6, 8)
3	(4, 1)	11	(4, 4)	19	(7, 8)
4	(5, 1)	12	(4, 5)	20	(8, 8)
5	(5, 2)	13	(4, 6)		
6	(6, 2)	14	(4, 7)		
7	(5, 2)	15	(4, 7)		

Task 5: Ghost Checkpoint ($t = 3$)

Additional rule: Lara must occupy cell $(4, 4)$ at exactly $t = 3$:

$$x_{3,4,4} = 1.$$

Result: Infeasible. BFS on the obstacle map (ignoring cameras) shows that the minimum number of steps to reach $(4, 4)$ from $(1, 1)$ is 6. The Manhattan distance alone is $|4-1|+|4-1| = 6$, and the obstacle cells $(4, 2)$ and $(4, 3)$ block shorter detours, making the actual minimum at least 6. Since the constraint requires arrival at $t = 3 < 6$, it is geometrically impossible regardless of the chosen path. The model has no feasible solution.

Extended Result (Teleport Allowed): If the movement constraint for the transition $t = 2 \rightarrow t = 3$ is relaxed (i.e. a "ghost teleport" is permitted), the model becomes feasible and yields $T^* = 14$ steps. Lara paces near the starting cell for the first two time steps, teleports to $(4, 4)$ at $t = 3$, then proceeds to the exit.

t	Cell	t	Cell	t	Cell
0	(1, 1)	5	(5, 3)	10	(8, 5)
1	(2, 1)	6	(6, 3)	11	(8, 5)
2	(1, 1)	7	(7, 3)	12	(8, 6)
3	(4, 4) *[teleport]*	8	(8, 3)	13	(8, 7)
4	(5, 4)	9	(8, 4)	14	(8, 8)

Task 6: Branch and Bound (Base Case)

Branching Strategy. If the LP relaxation gives a fractional solution, we branch on a fractional variable $x_{t,r,c}$ by creating two subproblems:

$$x_{t,r,c} = 0 \quad \text{and} \quad x_{t,r,c} = 1.$$

Bounding Strategy. At each node, we solve the LP relaxation.

- If the solution is integer → it is feasible for the IP and becomes the incumbent.
- If it cannot improve the current solution → the node is pruned.

Result for This Problem. Since $T^* = 14$ is fixed from BFS, the model becomes a feasibility problem. Due to tight constraints (movement and camera avoidance), the LP relaxation at the root node already yields a fully integer solution.

Search Tree.

Root Node LP relaxation solved Solution is integer ⇒ Optimal

Conclusion. No branching is required. The algorithm terminates at the root node, and the optimal

Question 2: CCI Production Planning

Problem Definition

Coca-Cola İçecek (CCI) plans production at two plants – Ankara (A) and Istanbul (I) – over $m = 1, \ldots, 6$ months. The objective is to minimize the total cost of inventory holding, cross-region transportation, workforce wages, hiring, layoffs, and bottling-mode switching, subject to demand, capacity, workforce, and cross-shipping constraints.

Data

Monthly demand $d_{j,m}$ (liters, L):

Region j	$m = 1$	$m = 2$	$m = 3$	$m = 4$	$m = 5$	$m = 6$
Western (W)	9,000,000	9,000,000	14,000,000	12,000,000	12,000,000	13,000,000
Central & Eastern (CE)	7,000,000	7,000,000	8,000,000	9,000,000	8,000,000	9,000,000

Plant parameters ($i \in \{A, I\}$):

Parameter	Ankara ($i = A$)	Istanbul ($i = I$)
Bottling rate, 2 L	6,000 bot/hr = 12,000 L/hr	9,000 bot/hr = 18,000 L/hr
Bottling rate, 1 L	10,000 bot/hr = 10,000 L/hr	15,000 bot/hr = 15,000 L/hr
Available hours/month	720	720
Initial workforce $w_{i,0}$	60	100
Primary region	CE	W

Cost parameters:

- Wage: $c_w = 1{,}200$ TL/employee/month
- Hiring: $c_h = 1{,}500$ TL/new employee
- Layoff: $c_l = 2{,}000$ TL/laid-off employee
- Inventory: $c_{\text{inv}} = 0.05$ TL/L/month
- Cross-region transport: $c_{\text{tr}} = 0.10$ TL/L
- Switching: $c_{\text{sw}} = 20{,}000$ TL/plant/month

Policy parameters: $\alpha = 0.25$, $\beta = 0.10$, $\Delta = 10$.

Note. The PDF states that 60% of demand is in 2 L bottles and 40% in 1 L bottles. Since the problem adds "*you may split accordingly*," we treat total demand as a single aggregate and let the optimizer choose the 1 L/2 L mix freely; this is already reflected in the bottling-rate constraint (D1).

Decision Variables

Indices: $i \in \{A, I\}$ plant, $j \in \{W, CE\}$ region, $m \in \{1, \ldots, 6\}$ month. All production and inventory quantities in liters (L).

- $p^1_{i,m},\ p^2_{i,m} \geq 0$ volume of 1 L / 2 L bottles produced at plant i in month m (L)
- $s_{i,j,m} \geq 0$ volume shipped from plant i to region j in month m (L)
- $I_{i,m} \geq 0$ end-of-month inventory at plant i in month m (L)
- $w_{i,m} \in \mathbb{Z}_{\geq 0}$ packaging workforce at plant i in month m
- $h_{i,m},\ l_{i,m} \in \mathbb{Z}_{\geq 0}$ employees hired / laid off
- $\delta_{i,k,m} \in \{0, 1\}$ 1 if bottle size $k \in \{1\text{L}, 2\text{L}\}$ is produced at plant i in month m
- $z_{i,m} \in \{0, 1\}$ 1 if both bottle sizes produced at plant i in month m

Objective Function

$$\min\ Z = C_{\text{inv}} + C_{\text{tr}} + C_w + C_h + C_l + C_{\text{sw}}$$

$$C_{\text{inv}} = 0.05 \sum_i \sum_m I_{i,m}$$

$$C_{\text{tr}} = 0.10 \sum_{i,j:j=\text{primary}(i)} \sum_m s_{i,j,m}$$

$$C_w = 1{,}200 \sum_i \sum_m w_{i,m}$$

$$C_h = 1{,}500 \sum_i \sum_m h_{i,m}$$

$$C_l = 2{,}000 \sum_i \sum_m l_{i,m}$$

$$C_{\text{sw}} = 20{,}000 \sum_i \sum_m z_{i,m}$$

Constraints

(D1) Bottling capacity (hours ≤ 720 per plant per month):

$$\frac{p^2_{A,m}}{12{,}000} + \frac{p^1_{A,m}}{10{,}000} \leq 720, \qquad \frac{p^2_{I,m}}{18{,}000} + \frac{p^1_{I,m}}{15{,}000} \leq 720 \qquad \forall\, m$$

(D2) Packaging capacity:

$$p^1_{i,m} + p^2_{i,m} \leq 100{,}000\, w_{i,m} \qquad \forall\, i, m$$

(D3) Workforce balance ($w_{A,0} = 60$, $w_{I,0} = 100$):

$$w_{i,m} = w_{i,m-1} + h_{i,m} - l_{i,m} \qquad \forall\, i, m$$

(D4) Workforce smoothing ($\Delta = 10$):

$$w_{i,m} - w_{i,m-1} \leq 10, \qquad w_{i,m-1} - w_{i,m} \leq 10 \qquad \forall\, i, m$$

(D5) Inventory balance (initial inventories = 0):

$$I_{i,m} = I_{i,m-1} + p^1_{i,m} + p^2_{i,m} - \sum_j s_{i,j,m} \qquad \forall\, i, m$$

(D6) Demand fulfillment (no backlogging):

$$\sum_i s_{i,j,m} \geq d_{j,m} \qquad \forall\, j, m$$

(D7) Cross-shipping cap (α = 25%):

$$s_{i,j,m} \leq \alpha\, d_{j,m} \qquad \forall\, i, m, j' = \text{primary}(i)$$

(D8) Safety stock (β = 10%, months $m = 1, \ldots, 5$):

$$I_{i,m} \geq \beta\, d_{j^*(i),\, m+1} \qquad m = 1, \ldots, 5 \quad \text{(quantities in L)}$$

where $j^*(i)$ is the primary region of plant i.

(D9) Zero ending inventory:

$$I_{i,6} = 0 \qquad \forall\, i$$

(D10) Switching cost (big-M = 50,000,000 L):

$$p^k_{i,m} \leq 50{,}000{,}000\, \delta_{i,k,m} \qquad \forall\, i, k, m$$

$$z_{i,m} \geq \delta_{i,1\mathrm{L},m} + \delta_{i,2\mathrm{L},m} - 1 \qquad \forall\, i, m$$